
Feature Prediction and Gait Laterality: A Geometry Audit of Skeleton JEPA

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Abstract

1 Human movement provides a useful test of whether predictive representations pre-
2 serve the geometry of an articulated body. Left-right reflection supplies a known
3 transformation, while differences between the two sides provide an observable
4 quantity whose sign should change under that transformation. We study this re-
5 lationship in a skeleton Joint-Embedding Predictive Architecture using pose se-
6 quences from gait videos. A source-separated evaluation compares trained and
7 initial encoders, tests reflection behavior, and changes the anatomical and tempo-
8 ral placement of hidden training targets. The available experiments show that pre-
9 dicting a clip’s hidden features more accurately than another clip’s features can
10 coexist with weaker recovery of a signed movement contrast. Motion-sensitive
11 summaries improve access to information already present at initialization, while
12 motion-weighted and connected-region masks provide no demonstrated advantage
13 over their matched random references. We distinguish this evidence from exact
14 symmetry imposed at the output and from synthetic demonstrations of explicit
15 reflection training. The study contributes an evaluation of articulated-body repre-
16 sentations for Physical World AI, identifying a gap between feature prediction and
17 the preservation of a task-relevant physical measurement.

18 1 Motivation and research question

19 Walking involves coordinated movement of an articulated body with corresponding left and right
20 limbs. This geometry provides both useful regularities and meaningful departures from them. An
21 average description of movement can remain unchanged when anatomical sides are exchanged,
22 whereas a quantity describing which side moves more should reverse sign. A representation in-
23 tended for physical prediction may need to support both responses.

24 The motivation differs across health conditions. Stroke can affect spatial and temporal gait sym-
25 metry, whereas Parkinson’s research also examines unequal arm swing. Cerebral palsy can involve
26 unilateral or bilateral impairments, and muscular disorders may alter pelvic and limb coordination.
27 These examples make anatomical correspondence useful, while warning against reducing abnormal
28 gait to one scalar.

29 These are literature-based motivations, not findings about clinical differences in our cohort. In
30 particular, the broad myopathic annotation does not identify Duchenne muscular dystrophy. The data
31 do not supply verified affected-side labels, standardized disease severity, or the subtype information
32 needed to connect target sign to a particular impairment. Here, laterality means a signed comparison
33 of estimated left- and right-side movement.

34 We ask whether masked feature prediction makes that comparison easier to recover from a skele-
35 ton encoder on held-out source videos. The initial null is that training provides no improvement
36 over the same encoder at initialization under a fixed downstream evaluation. Separate questions

Local annotation	Clips / sources	Relevant published observation and implication
Normal	270 / 29	Supplies the dataset reference category; a zero signed contrast is not a clinical definition of normal walking.
Stroke	75 / 18	Spatial and temporal asymmetries differ across walkers; a single speed contrast covers only part of gait behavior. Patterson et al. [1]
Parkinson’s	39 / 9	Arm-swing asymmetry motivates preserving side identity; our shoulder points do not measure arm swing directly. Lewek et al. [2]
Cerebral palsy	58 / 9	Studies include unilateral and bilateral presentations and examine speed-dependent symmetry. Gait-speed study [3]
Myopathic	183 / 28	Dystrophy studies examine pelvic compensation and multisegment asymmetry, illustrating features beyond this target. Longitudinal kinematics [4], kinematic-synergy study [5]

37 concern whether the output changes sign correctly and whether landmark features follow a specified
38 reflection rule. Answering one does not establish the others.

39 Our contribution is a controlled evaluation of a small predictive representation model. The com-
40 pleted experiments concern whole-clip masked prediction and frozen-feature readouts. Their rele-
41 vance to world models is the assessment of a geometric property that could matter to a future dy-
42 namics model. The study has no demonstrated real-data forecasting, planning, or multimodal-fusion
43 result.

44 2 Data and the geometric target

45 The local subset of the Gait Abnormality in Video Dataset [6] contains 666 annotations, of which
46 642 have pose archives. Quality control retains 625 sequences from 93 source videos. The anno-
47 tations cover normal, Parkinson’s, stroke, myopathic, and cerebral-palsy gait. They stratify source
48 assignments and describe the dataset; they do not supervise encoder training.

49 Each archive contains a variable-length sequence of 33 landmarks with estimated coordinates and
50 visibility. Coordinates use image-normalized horizontal and vertical positions and inferred depth
51 scaled relative to crop width. Their Euclidean differences are coordinate-derived motion measures,
52 not calibrated metric three-dimensional speeds. Aspect ratio, projection, pose-estimation errors and
53 visibility can affect them.

54 For each of five bilateral pairs—shoulders, knees, ankles, heels, and foot tips—we calculate left and
55 right speeds using observed timestamps and only transitions for which both landmarks are observed
56 at both ends. Let their median speeds be $m_{L,k}$ and $m_{R,k}$. The target is

$$y(x) = \frac{1}{5} \sum_{k=1}^5 \frac{m_{L,k} - m_{R,k}}{m_{L,k} + m_{R,k} + \epsilon}.$$

57 All five pairs must provide at least eight shared valid transitions. Hips establish the pelvis reference
58 but are excluded from this target: pelvis centering makes their speed magnitudes identical. The target
59 is calculated before input interpolation and resizing. It measures asymmetry of median coordinate
60 speed, not stride duration, phase coordination, or limb loading. Oppositely signed pair contrasts can
61 cancel in the average. Similarly, left- and right-affected people could have substantial individual
62 asymmetry but a near-zero group-average signed target. A small y therefore does not establish
63 healthy or bilaterally symmetric gait.

64 Reflection M negates the centered horizontal coordinate and exchanges all anatomically paired land-
65 marks, together with validity. Consequently, $M^2x = x$ and $y(Mx) = -y(x)$. This identity is a
66 property of the coordinate operation and formula, rather than evidence that every reflected recording
67 would occur naturally.

68 The encoder input is prepared separately. Visibility at least 0.45 and finite coordinates determine
69 observations; up to four missing samples can be interpolated for input only. Pelvis normalization
70 removes translation and applies a body scale. Resizing then samples 64 locations along normalized

sequence index, without preserving the original timestamp intervals. Values under invalid entries are zero-filled only alongside an explicit validity mask.

Stage	Array shape	What remains associated; what changes
Archived clip	$T_i \times 33 \times 4$, plus frame numbers and frame rate	Coordinate channels and visibility remain linked to one source and sequence.
Target branch	$T_i \times 33 \times 3$, validity	Original timestamps and observed-only paired transitions determine the target.
Model input	$64 \times 33 \times 3$, validity	Joint identities remain; interpolation, normalization and resampling alter coordinate values and timing.
Four-frame patches	64×33	Each patch concatenates four coordinates for the same landmark; complete validity determines eligibility.
Embedded token grid	$16 \times 33 \times 12$	Time-block and landmark indices identify each 96-dimensional feature.
Batch with mask	$B \times 528 \times 96$, plus target indices and validity	Gathering and padding retain each clip’s own targets; paired arms match counts per clip, which may vary across the batch.

The pipeline preserves the correspondence between recordings, landmarks, masks, and targets. It cannot promise preservation of the exact observed shape or speed after preparation. Cohort preparation occurs before split-file creation in the implementation; it uses deterministic clip-local operations rather than statistics fitted across sources. Source membership is fixed before training sampling, augmentation, and model fitting.

3 Training and source separation

All clips and derived views from a source video inherit its fold. Five outer folds contain respectively 19, 19, 19, 18, and 18 test sources; their test sequence counts are 189, 182, 72, 77, and 105. Each encoder trains only on its matching outer-training sources. New encoders are initialized for each fold and seed, and source-balanced sampling limits domination by videos with many clips.

The base model uses a 96-dimensional embedding, four encoder layers, two predictor layers, and four attention heads. The online encoder receives a full token grid with hidden coordinate content suppressed and positional information retained; a predictor estimates teacher features at hidden positions. A target encoder supplies those features and is updated by an exponential moving average (EMA) of online weights. Its parameters receive no gradients.

The real-data objective matches distributions over feature channels using cross-entropy,

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(p_{\bar{\theta}}, p_{\theta, \phi}) + 0.05 \mathcal{L}_{\text{VICReg}}.$$

Here $\theta, \phi, \bar{\theta}$ denote online-encoder, predictor, and teacher parameters. Teacher centering and separate temperatures form the target and predicted distributions. VICReg penalizes insufficient feature variation and redundant channels, alongside agreement between geometric views. It is calculated from unmasked views pooled over twelve gait landmarks. Consequently, anatomical information remains in the regularizer even when targets are chosen across the body. Mean-squared feature error appears in diagnostics; it is not the real-data pretraining objective.

The historical base profile records 300 epochs, four updates per epoch, AdamW, batch size 20, learning rate 10^{-3} , and weight decay 0.05. All completed controlled grids use 1,200 updates per encoder. Mask fraction is applied to eligible gait tokens and reduced to a feasible shared count; the original 0.6 setting does not hide 60% of all 528 tokens.

The study introduces anatomical laterality at different locations that should be kept explicit:

The latest motion/region trainer fixes reflection probability and explicit symmetry weight to zero. It does not test a combined motion-plus-reflection recipe. None of these encoder objectives receives y , a diagnosis, or an affected-side label.

After pretraining, encoder weights are frozen. Inner source-separated validation selects a ridge-regression penalty; ridge regression is a linear readout whose weight penalty limits fitting unstable feature combinations. The original audit uses four inner folds, while later comparisons use three.

Entry point	Effect on training or evaluation	Evidence status
Named left/right landmarks and gait pooling	Preserve anatomical identities and choose locations contributing to regularization	Used in completed real-data training
Gait-only target eligibility	Select hidden tokens from six bilateral landmark pairs while retaining all-body context	Historical base and Notebook 12 gait arm
Anatomical reflection augmentation	Exchange joint identities, horizontal signs, validity and transformed views consistently	Previously reported original variant
Explicit reflection penalty	Penalize token disagreement after anatomical exchange; gradients update the online encoder	Notebook 09 synthetic demonstration
Odd readout and signed regression target	Constrain or evaluate a frozen representation	Downstream only

Outer-training groups contain 74 or 75 sources. Each original inner fit contains 55–57 sources, and validation contains 18 or 19. No outer-test source enters these fits. Imputation and scaling are fitted within each inner fitting partition and refitted on the outer-training set after selection. These inner folds select the readout only: the encoder has already seen their unlabeled inputs during outer-training pretraining.

4 Evaluation through successive questions

4.1 Does the prepared input preserve the measurement?

The original audit prompted a check before attributing weak performance to the encoder. Notebook 07 recomputes the movement formula on the prepared input. Repeating that descriptive calculation during this review produces finite pairs for 623 clips from 92 sources. Source-weighted sign agreement with the original target is 70.4%; the direct calculation-agreement R^2 is 0.218. Two clips cannot supply a finite recomputation.

This comparison changes the entire preparation path, including timing, interpolation and valid support. It identifies a measurement mismatch without localizing its cause. Its R^2 is neither a model score nor an information ceiling: a learned mapping from prepared input could behave differently. It motivates evaluating the target after individual preparation stages on the same eligible clips.

4.2 Does the original representation satisfy the reflection audit?

The retained original aggregate summary reports source-balanced $R^2 = 0.060$, with a 95% source-bootstrap interval of $[-0.025, 0.126]$. The learned-minus-initial difference is -0.018 , with interval $[-0.039, 0.002]$. Reflection augmentation changes R^2 by 0.004, with interval $[-0.006, 0.013]$. These estimates do not demonstrate a training benefit or establish equivalence between variants.

The direct token test aligns landmark positions under reflection while leaving feature channels unchanged. Its normalized squared discrepancy is

$$q = \frac{\|Z(Mx) - SZ(x)\|_C^2}{\|Z(Mx)\|_C^2 + \|SZ(x)\|_C^2},$$

where S exchanges joint identities and C selects common valid tokens. The retained summary reports $q = 0.114$ for learned features and 0.083 at initialization, with a paired increase of 0.031, interval $[0.016, 0.048]$. This is unfavorable evidence for the specified identity-channel transformation. An encoder could express reflection through a different transformation of feature channels; the test does not examine that possibility.

An output can be made antisymmetric through an odd feature, $z^-(x) = [z(x) - z(Mx)]/\sqrt{2}$, followed by a linear readout without an intercept or feature centering. Its symmetry follows algebraically. Predictive accuracy still requires evidence: the recorded learned-minus-initial difference under this construction is negative. A nonzero constant feature vector can also have zero token discrepancy while conveying no clip-specific information. Rejecting zero-energy tensors alone does

not exclude that case, which is why the geometry tests must be paired with feature-variation and predictive controls.

The original values are available in a retained figure-summary JSON. Its full report, checkpoint, and per-example prediction chain was not found in this checkout during the present review. We retain these results as previously reported context and distinguish them from the newly audited comparisons below.

4.3 Does anatomical target selection improve learning?

Notebook 12 compares scattered hidden targets selected from twelve gait landmarks with the same number selected across all 33 landmarks. Both encoders still receive the full landmark set and retain anatomical choices in regularization and readout. Starting weights, sampled clips, geometric views, and 1,200 updates are paired.

The teacher-feature readouts obtain $R^2 = -0.023$ with gait targets and -0.010 with all-landmark targets; the initial encoder obtains 0.048. The broader-minus-gait difference has interval $[-0.018, 0.050]$, leaving both improvement and harm compatible with these recordings. Larger ridge penalties frequently reach the edge of the search range, limiting conclusions about the best achievable readout. The later grid extends the candidates to 10,000, which is still selected by 39% of teacher motion-sensitive fits and 77% of the corresponding online fits. The reported deficits are results of these readout procedures, rather than a bound on what any decoder could recover.

Notebook 08’s earlier values are retained in the tutorial but excluded from the main comparison: implementation and readout choices changed together before Notebook 12. Likewise, Notebook 12’s direct-pose $R^2 = 0.130$ and the latest 0.0345 come from different evaluation recipes. Their change cannot be attributed to a mask or a single regularization choice.

4.4 Does targeting motion or connected anatomy help?

Notebooks 15–18 extend the comparison to motion-weighted and connected-region targets. The completed grid contains 125 trained encoders, each with 1,200 updates. Motion comparisons and region comparisons have their own random references, matched to the actual hidden count. Equal nominal mask percentages would not ensure equal tasks because missing observations and eligible landmark pools differ.

Frozen features or control	Mean $R^2 \pm$ seed SD	Mean absolute error
Training-source target mean	-0.0106 ± 0.0000	0.046172
Direct prepared-pose summary	0.0345 ± 0.0000	0.044960
Initial encoder, mean summary	0.0708 ± 0.0186	0.044344
Initial encoder, motion-sensitive summary	0.2225 ± 0.0268	0.041546
Motion-uniform teacher, motion-sensitive	0.1137 ± 0.0110	0.043658
MAMP-style teacher, motion-sensitive	0.1129 ± 0.0306	0.043751
Robust-motion teacher, motion-sensitive	0.1142 ± 0.0210	0.043595
Region-uniform teacher, motion-sensitive	0.1094 ± 0.0088	0.044037
Connected-region teacher, motion-sensitive	0.1008 ± 0.0092	0.043756

Each row covers 625 clips and 93 sources. SD describes the five optimization seeds, not sampling uncertainty. The initial encoder, direct-pose, and training-mean controls are reused across arms; their repeated appearance in machine-readable tables is not independent replication. “Initial” refers only to the S-JEPA weights: the pipeline still uses a pretrained pose detector, supplied anatomical identities, feature engineering, and a supervised ridge readout.

The motion-sensitive summary adds temporal variation, feature changes, and observation support, increasing the readout from 960 to 2,890 features. All trained arms remain below their matched initial encoder in each of the five seeds. Its advantage over mean pooling cannot yet be attributed solely to motion because the summary also includes support information. Temporal standard deviation is insensitive to ordering. The initial gain is 0.1517 in R^2 , compared with 0.0270–0.0468 for trained teachers. The online encoders also underperform initialization, with mean R^2 between 0.0636 and 0.0809, so evaluating the teacher alone does not explain the deficit.

We additionally recomputed trained-minus-initial intervals from the saved predictions for this revision. They are new exploratory analyses, using 2,000 paired source resamples with bootstrap seed 812 and no model retraining. All five intervals lie below zero; the motion-uniform contrast is -0.1088 , interval $[-0.1701, -0.0412]$. The full set appears in Appendix A. This strengthens the conditional description of a learning deficit while leaving evaluation choices and adaptive development as limitations.

Mask minus its matched random reference	ΔR^2	95% source interval
MAMP-style motion weighting	-0.0008	$[-0.0204, 0.0153]$
Robust motion mixture	0.0005	$[-0.0155, 0.0152]$
Connected region	-0.0087	$[-0.0333, 0.0177]$

The mask audits confirm that the proposed changes took effect. Motion targets average 80.771 valid tokens per clip, approximately 17% of valid all-body tokens. Relative to eligible-token motion, average enrichment is approximately zero for random selection, 0.0175 for MAMP-style weighting, and 0.0378 for the robust mixture. The mixture retains a uniform component and limits extreme motion scores; high displacement may still reflect tracking error.

In the structure audit, immediate visible time brackets surround 69.9% of random-reference targets and none of the connected-region targets. Connected regions also reduce visible anatomical-neighbor availability from 0.988 to 0.512 and hide about 9.9% of valid tokens. Thus the masks remove different observable clues, although they do not isolate what the contextualized teacher encodes. Whole trajectories and interior gaps are audited for coverage, but are absent from the completed training comparison.

These comparisons are development analyses on the reused GAVD cohort. The motion and region families use different hidden budgets and separate random references. We compare each alternative only with its own reference. The largest numerical score among arms is not a validated best mask, and an interval spanning zero does not establish equal performance.

4.5 What does successful feature prediction establish?

The trained predictors consistently predict their own clip’s hidden teacher features more accurately than targets from another source video, comparing matched and mismatched errors on the same eligible control clips and target positions. Initial predictors show no consistent preference. This establishes useful clip correspondence for that diagnostic. Posture, viewpoint, observation support, or motion could contribute to it; the result does not isolate a physical mechanism.

Thus, predictor training can succeed on its own features while the frozen representation yields weaker laterality readouts. Because each arm supplies its own changing teacher, raw feature-prediction losses are not a common between-model measurement scale. The coordinate-derived outcome supplies a shared evaluation target.

5 Statistical interpretation

For V sources and n_v clips from source v , each clip has weight $w_{vi} = 1/(Vn_v)$. With $\bar{y}_w = \sum w_{vi} y_{vi}$,

$$R_w^2 = 1 - \frac{\sum_{vi} w_{vi} (y_{vi} - \hat{y}_{vi})^2}{\sum_{vi} w_{vi} (y_{vi} - \bar{y}_w)^2}.$$

Each source therefore receives equal total weight even when it supplies many clips. Within a seed, predictions from all five held-out folds are pooled before computing R^2 ; the reported result averages the five seed-specific scores. It is neither an average of fold R^2 values nor a score from an ensemble of seed-averaged predictions. Negative R^2 means squared error exceeds variation around the weighted evaluation mean; a separately fitted training-source mean is the usable prediction baseline.

Paired uncertainty resamples entire videos while retaining their clips and paired seed predictions. For a mask comparison the null is no positive difference in source-balanced readout performance relative to the count-matched random policy; for learning it is no positive difference relative to matched initialization. These are superiority questions, and no equivalence margin or power guarantee is

222 established. The latest mask contrasts and this revision’s learning contrasts use 2,000 resamples.
223 Their intervals condition on the fitted models and exclude full-retraining and new-split uncertainty.
224 The historical original summary records a separate 2,000-resample analysis, whose complete inputs
225 remain unavailable. Repeated seeds and diagnostic mask draws are not independent participants.
226 Subsequent experiments were informed by the same development cohort, so their source separation
227 does not make the overall research trajectory an untouched confirmatory study.

228 6 Implications and limitations

229 The strongest supported conclusion concerns access to one signed movement measurement under
230 the evaluated readouts. A negative learned-over-initial contrast does not demonstrate that all mo-
231 tion information has disappeared. Mean pooling, feature scaling, regularization, and the difference
232 between the original target and prepared input remain possible contributors.

233 The study fits the workshop’s articulated geometry and evaluation themes. Coordinates derived
234 from RGB do not constitute multimodal sensing, and an inferred depth coordinate does not estab-
235 lish calibrated 3D reconstruction. Clinical interpretation would require independent measurements,
236 affected-side labels, and participant-separated validation. The source-video split cannot exclude the
237 same unidentified person appearing in different videos.

238 The next experiments should first isolate support features, broaden training-only ridge selection,
239 and measure the target after each preparation stage. A later study can test whether explicit geo-
240 metric training improves common observable future endpoints, using past-only preprocessing and
241 persistence and velocity baselines. Existing symmetry-training and future-decoding demonstrations
242 are synthetic; they are not evidence of forecasting gait recordings.

243 Institutional ethics, data-use, and derived-pose release reviews remain unresolved in the recorded
244 governance status. This manuscript is an internal revision, and it does not claim those determinations
245 have been obtained.

246 7 Related work and the remaining empirical question

247 Masked feature prediction in I-JEPA [7] and S-JEPA [8] motivates the model family. MAMP [9]
248 motivates motion-informed targets and sampling, while SLiM [10] provides recent connected-region
249 masking precedent. Our MAMP-style arm adapts its official-code sampling convention while retain-
250 ing the JEPA feature objective; it is not a reproduction of MAMP’s complete motion-prediction
251 method.

252 seq-JEPA [11] studies invariant and equivariant representations. Soft Equivariance Regulariza-
253 tion [12] examines geometric regularization and its tradeoffs. Our separation of a geometric score
254 from predictive utility belongs in that established context.

255 Recent preprints make the next comparison more specific. V-JEPA 2.1 [13] studies dense and mul-
256 tilayer predictive supervision; Human-JEPA [14] examines human-focused perception and anticipa-
257 tion. GaitJEPA [15] already applies predictive learning to gait silhouettes. These works limit any
258 claim of novelty based on combining gait with JEPA. They motivate evaluating what a predictor
259 adds beyond its encoder on a shared movement endpoint, without predicting a gain on this small
260 pose dataset.

261 Our distinct empirical question is whether successful clip-feature correspondence and deliberately
262 altered target geometry improve access to an observed signed movement quantity. The current evi-
263 dence supports a negative result under the evaluated training and readout procedures. A transferable
264 mechanism or a verified remedy remains to be established.

265 8 Reproducibility

266 The tutorial, source-split reference, core audit, and extension audit document the evidence and its
267 limits. Existing notebooks and artifacts were left unchanged. The absent original report chain
268 remains unavailable; the later prediction grids and the newly calculated exploratory intervals have
269 separate provenance. This revision adds no training run or clinical finding.

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A Additional exploratory learning contrasts

These contrasts were calculated during the 9 September revision from the complete latest prediction grid. They were not part of the original registered reflection audit. Each compares the final teacher’s motion-sensitive summary with its matched initial encoder using the same readout family, clips and seed. Intervals are marginal percentile source-bootstrap intervals conditional on the fitted models; there is no family-wise multiplicity adjustment.

Training arm	Learned minus initial R^2	95% source interval
Motion uniform	−0.1088	[−0.1701, −0.0412]
MAMP-style motion	−0.1097	[−0.1651, −0.0514]
Robust motion	−0.1083	[−0.1706, −0.0423]
Region uniform	−0.1131	[−0.1662, −0.0596]
Connected region	−0.1218	[−0.1828, −0.0553]

The recomputation script checks the prediction coverage and reproduces the recorded pooled scores before calculating these contrasts. The extension evidence audit records the retained artifact paths and full-precision values.

B Competing explanations and tests that could distinguish them

Explanation compatible with the findings	Useful next test	Outcome that would weaken it
Preparation changes the relevant measurement	Recompute the target after each preparation stage on common valid transitions, with timestamps retained	Little discrepancy at every stage despite poor learned readouts
Summary choice obscures useful features	Compare mean, support-only, temporal SD, increments and their combinations on frozen initial and trained encoders	Deficit persists under adequately selected readouts with matched capacity
Ridge regularization is insufficient	Expand the penalty grid identically for all arms using inner training-source validation	Interior selected optima still produce a learned deficit
The prescribed reflection action is too restrictive	Fit a shared involutive channel map on training pairs and evaluate held-out pairs, with initial and mismatched controls	No meaningful advantage for genuine reflected pairs on held-out sources
Predictive targets emphasize another clip property	Compare local motion supervision or visible-token targets at matched budgets and measure a common observable outcome	Improved internal correspondence still fails to improve the common outcome

These are proposed discriminating tests, not explanations established by the current data. Fitting a reflection map that preserves lengths would add an orthogonality assumption; the input reflection does not guarantee that learned feature channels obey it. Any new decision based on this cohort remains development work and should precede confirmation in an independently specified setting.